L273 IS AUTO COIL MACHINE

OPERATION MANUAL

MACHINE FRONT PANEL

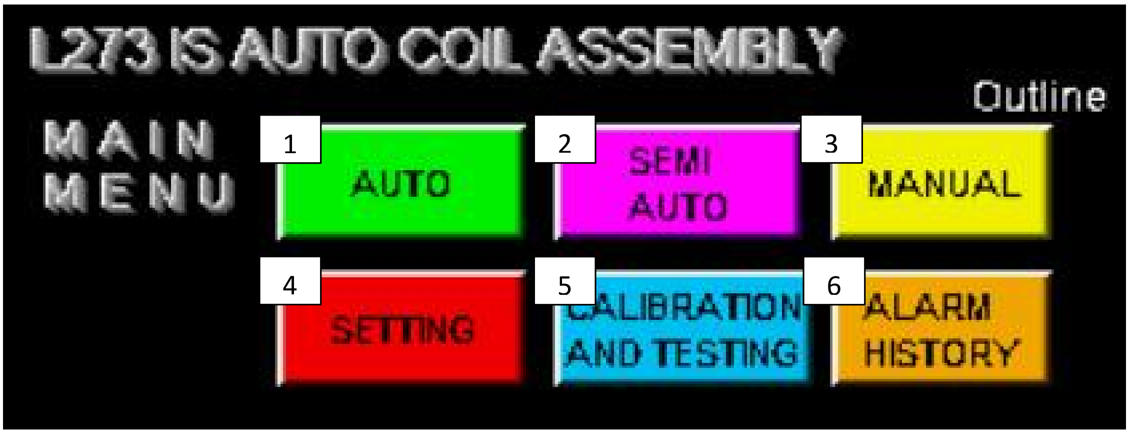


Figure 1 : Main panel

MAIN PANEL

No	Function	Description
1	AUTO MODE	WHEN TURNED ON, THE MACHINE IS IN AUTO MODE AND READY FOR PRODUCTION. WHEN TURNED OFF, THE MACHINE IS IN DEBUGGING OR REPAIRING MODE.
2	SEMI AUTO MODE	TO ENTER SEMI AUTO MODE
3	MANUAL	TO ENTER IO & SERVO SETTING MODE
4	SETTING	TO ENTER SETTING MODE
5	CALIBRATION AND TESTING	TO ENTER DRIVER CALIBRATION MODE
6	ALARM HISTORY	TO CHECK ALARM HISTORY

MACHINE OPERATION PANEL

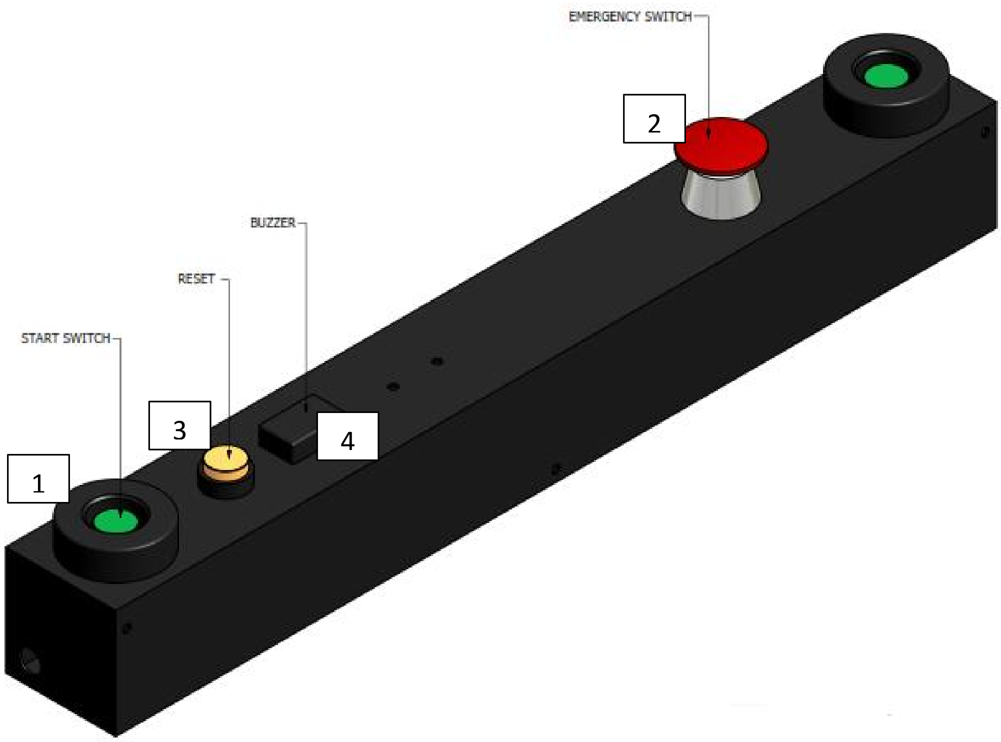


Figure 2 : Operation panel

OPERATION PANEL

No	Function	Description
1	START BUTTON	PRESS TO START THE MACHINE
2	EMERGENCY BUTTON	PRESS TO EMERGENCY STOP
3	RESET BUTTON	PRESS TO RESET THE MACHINE AND BUZZER
4	BUZZER	WILL TURN ON WHEN PROCESS COMPLETE OR EMERGENCY STOP

MACHINE OPERATION PROCEDURE

1. POWER ON THE MACHINE BY SWITCH ON POWER BUTTON.

2. TURN ON AIR SUPPLY TO THE MACHINE AND MAKE SURE EMERGENCY BUTTON IS IN OFF POSITION.

3. PRESS RESET BUTTON AND MACHINE WILL PERFORM HOMING ROUTINE.

4. GOTO SETTING MODE AND SELECT SLIDER SPEED.

5. SET MACHINE TO AUTO MODE.

6. LOAD PRODUCT IN MACHINE AND PRESS THE START BUTTON.

7. WHEN PROCESS IS COMPLETED, THE BUZZER WILL SOUND 3 TIMES AND OPERATOR CAN UNLOAD THE PRODUCT FROM THE MACHINE.

8. REPEAT PROCESS 5 AND 6 TO CONTINUE PRODUCTION.

SETTING FUNCTION

Figure 3 : Speed Screen

SPEED SETTING SCREEN

No	Buttons	Description
1	120%	Machine run at 120% of normal speed
2	100%	Machine run at 10% of normal speed
3	80%	Machine run at 80% of normal speed
4	BACK	Back to previous page
5	NEXT	Go to next page
6	MENU	Go to Menu Page

Figure 4 : Semi Auto Screen

SEMI AUTO SCREEN

No	Function	Description
1	LOAD	To start the tray load sequence
2	UNLOAD	To start the tray unload sequence
3	GLUE	To start the glue sequence
4	UV	To start the UV sequence
5	COIL 1 PICK & COIL 1 PLACE	To start the coil 1 load sequence at selected no
6	COIL 2 PICK & COIL 2 PLACE	To start the coil 2 load sequence at selected no

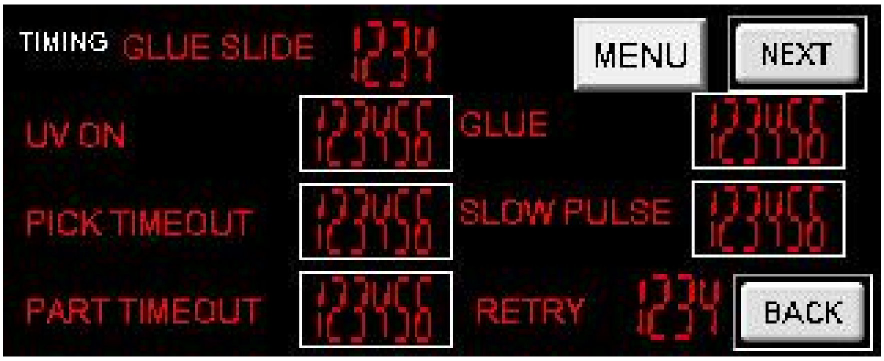


Figure 5 : Timing Screen

TIMING SCREEN

No	Function	Description
1	GLUE SLIDE	Timing for glue slide ON
2	UV ON	Timing for UV ON
3	PICK TIMEOUT	Timeout for coil pick
4	PART TIMEOUT	Timeout for base/part pick
5	GLUE	Timing for GLUE
6	SLOW PULSE	Speed change for Z Axis point
	RETRY	Retry for coil pick

Figure 6 : Servo Speed Screen

SERVO SPEED SCREEN

No	Function	Description
1	ACC/DEC	Acc/Decc speed for each of the axis
2	OPERATING	Operating speed for each of the axis

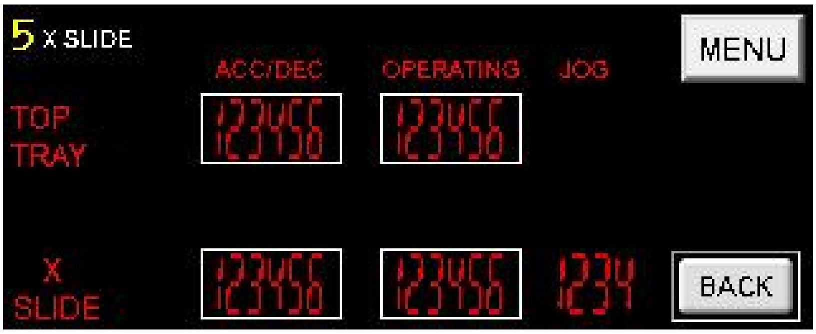


Figure 7 : Servo Speed Screen

SERVO SPEED SCREEN

No	Function	Description
1	ACC/DEC	Acc/Decc speed for each of the axis
2	OPERATING	Operating speed for each of the axis

SEMI AUTO FUNCTION

Figure 8 : Semi Auto Screen

SEMI AUTO SCREEN

No	Function	Description
1	COIL 1 PICK	Pick Coil 1 process will be turned on when selected
2	COIL 1 PLACE	Place Coil 1 process will be turned on when selected
3	COIL 2 PICK	Pick Coil 1 process will be turned on when selected
4	COIL 2 PLACE	Place Coil 2 process will be turned on when selected
5	GLUE 1	Apply Glue 1 process will be turned on when selected
6	GLUE 2	Apply Glue 2 process will be turned on when selected
7	GLUE 3	Apply Glue 3 process will be turned on when selected
8	GLUE 4	Apply Glue 4 process will be turned on when selected
9	GLUE 5	Apply Glue 5 process will be turned on when selected
10	UV 1	UV 1 process will be turned on when selected
11	UV 2	UV 2 process will be turned on when selected

MANUAL FUNCTION

Figure 9 : Semi Auto Screen

MANUAL SCREEN

No	Function	Description
1	I/O CONTROL	Go to I/O Control screen
2	SERVO CONTROL	Go to Servo Control screen

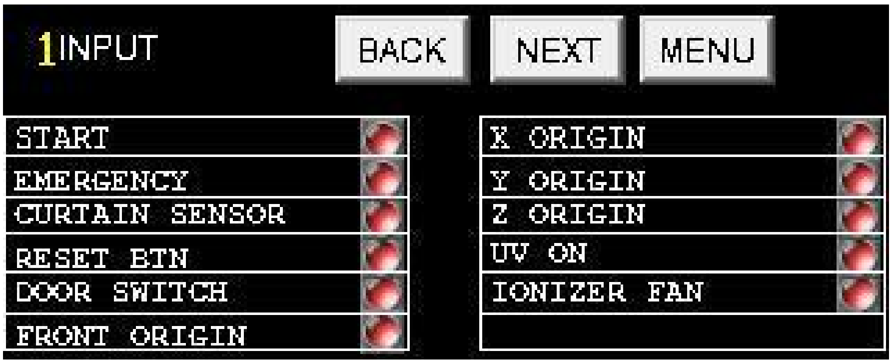


Figure 10 : Input Screen

INPUT SCREEN

No	Function	Description
1	INPUT	Show status of the input signals

Figure 11 : Input Screen

INPUT SCREEN

No	Function	Description
1	INPUT	Show status of the input signals

Figure 12 : Output Screen

OUTPUT SCREEN

No	Function	Description
1	OUTPUT	Menu to trigger all output signals

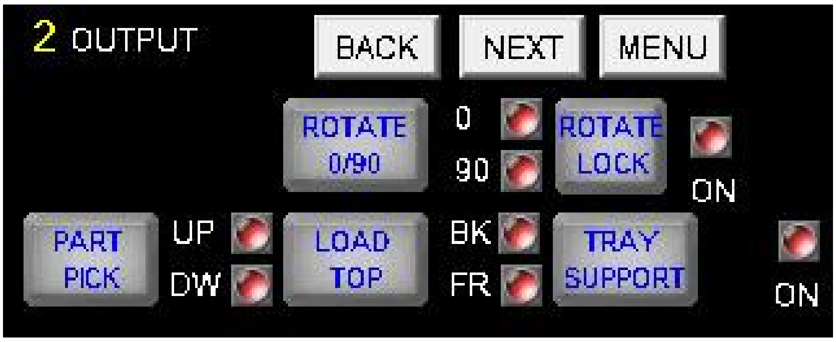


Figure 13 : Output Screen

OUTPUT SCREEN

No	Function	Description
1	OUTPUT	Menu to trigger all output signals

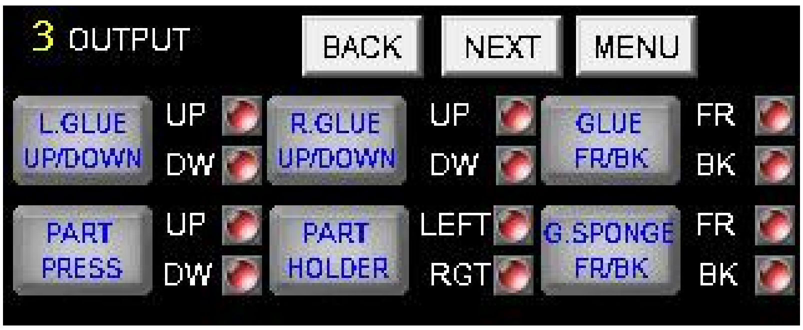


Figure 14 : Output Screen

OUTPUT SCREEN

No	Function	Description
1	OUTPUT	Menu to trigger all output signals

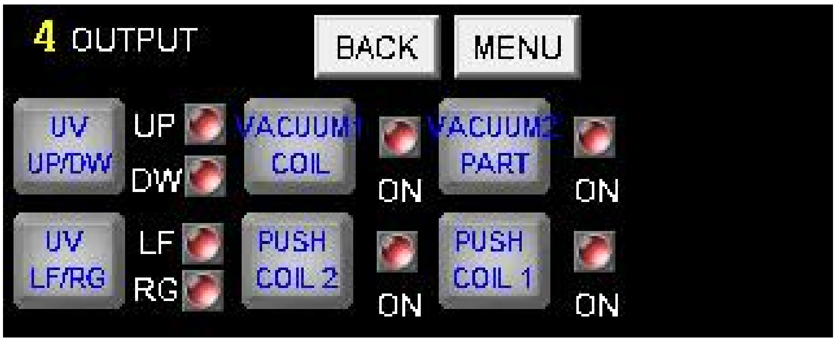


Figure 15 : Output Screen

OUTPUT SCREEN

No	Function	Description
1	OUTPUT	Menu to trigger all output signals

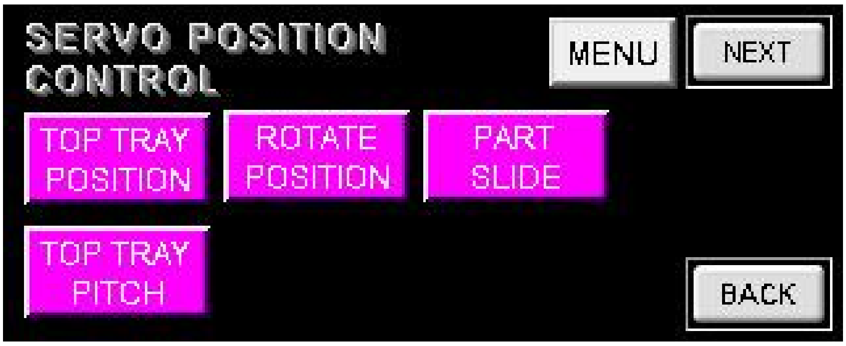


Figure 16 : Servo Control Menu Screen

SERVO CONTROL SCREEN

No	Function	Description
1	TOP TRAY POSITION	Go to screen to teach coil pick up position
2	ROTATE POSITION	Go to screen to teach tray drop position
3	PART SLIDE	Go to screen to teach part carrier slide position
4	TOP TRAY PITCH	Go to screen to setup coil tray dimensions & servo calculation specs

Figure 17 : Servo Control Menu Screen

SERVO CONTROL SCREEN

No	Function	Description
1	COIL1 PLACING	Go to screen to teach coil pick up position
2	COIL2 PLACING	Go to screen to teach coil pick up position
3	OTHER POS	Go to screen to teach other position

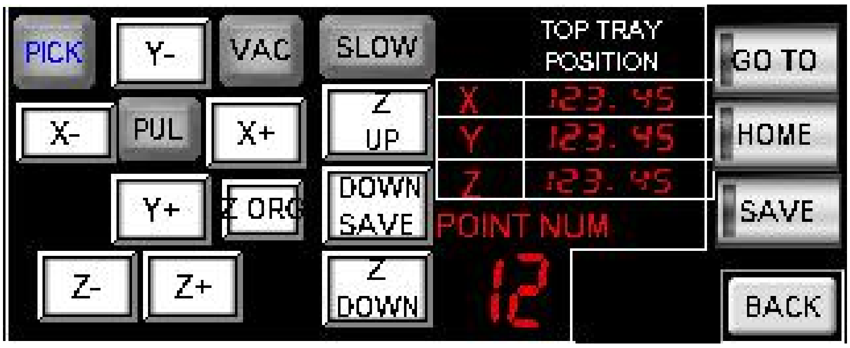


Figure 18 : Top Tray Position Screen

TOP TRAY SCREEN

No	Function	Description
1	X+	Move X-Axis to the right
2	X -	Move X-Axis to the left
3	Y+	Move Y-Axis to the frontward
4	Y-	Move Y-Axis to the backward
5	Z+	Move Z-Axis to the downward
6	Z-	Move Z-Axis to the upward
7	Z UP	Move Z-Axis to Z origin point
8	Z DOWN	Move Z-Axis to selected point no.
9	DOWN SAVE	Save Z-Axis to selected point no.
10	SLOW	Make jog slow or fast
11	PICK	Turn on pick cylinder
12	PUL	Turn on pulse

13	VAC	Turn on vacuum
14	GOTO	Move X & Y Axis to selected point no.
15	HOME	Move X,Y & Z Axis to Origin position
16	SAVE	Save X-Axis & Y-Axis to selected point no.

Figure 19 :Rotate Position Screen

ROTATE TRAY SCREEN

No	Function	Description
1	UP	Move Rotate Axis Up
2	DW	Move Rotate Axis Down
3	SLOW	Make jog slow or fast
4	TRAY SUPPORT	Turn on bottom tray support
5	GOTO	Move Rotate Axis to selected point no.
6	HOME	Move Rotate Axis to Origin position
7	SAVE	Save Rotate Axis to selected point no.

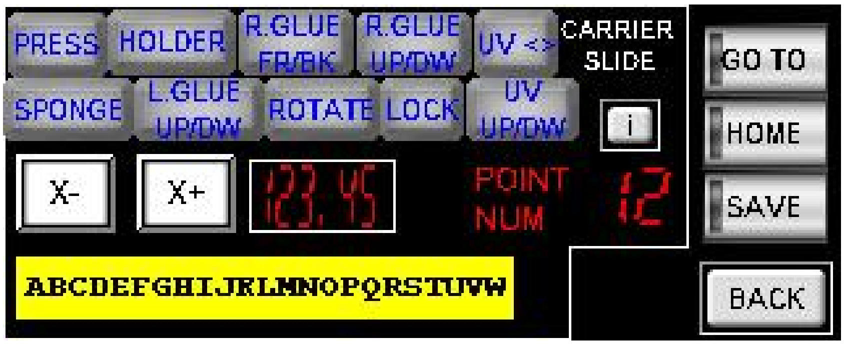


Figure 20 :Rotate Position Screen

PART CARRIER SLIDE SCREEN

No	Function	Description
1	X+	Move Part Carrier Slide to the right
2	X-	Move Part Carrier Slide to the left
3	i	Show information of all the point no.

4	GOTO	Move Part Carrier Axis to selected point no.
5	HOME	Move Part Carrier Axis to Origin position
6	SAVE	Save Part Carrier Axis to selected point no.
7	“output signal”	PRESS,HOLDER,R.GLUE_FR/BK,R.GLUE_UP/DW,UV<>, SPONGE,L.GLUE_UP/DW,ROTATE,LOCKUV_UP/DW.

Figure 21 : Top Tray Pitch Setting Screen

TOP TRAY PICTH SCREEN

No	Function	Description
1	X PITCH	Top Tray(Coil Tray) X pitch
2	Y PITCH	Top Tray(Coil Tray) Y pitch
3	ROW	The number of coil rows
4	COL	The number of coil columns
5	TOTAL	Total number of coils (auto calculated)
6	1mm(X)	Servo X pulse specs
7	1mm(Y)	Servo Y pulse specs

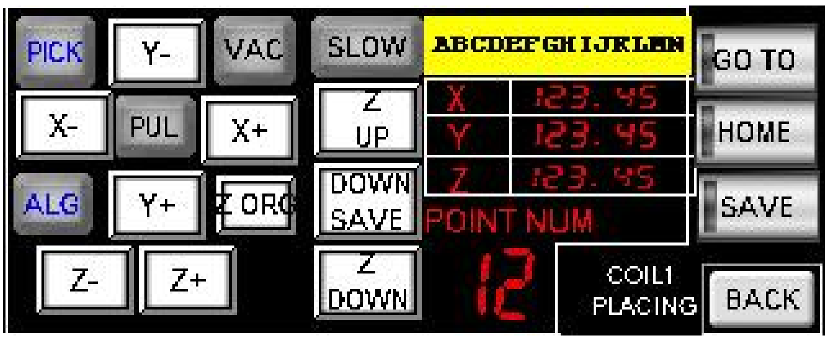


Figure 22 : Coil 1 Placing Screen

COIL 1 PLACING SCREEN

No	Function	Description
1	X+	Move X-Axis to the right
2	X -	Move X-Axis to the left

3	Y+	Move Y-Axis to the frontward
4	Y-	Move Y-Axis to the backward
5	Z+	Move Z-Axis to the downward
6	Z-	Move Z-Axis to the upward
7	Z UP	Move Z-Axis to Z origin point
8	Z DOWN	Move Z-Axis to selected point no.
9	DOWN SAVE	Save Z-Axis to selected point no.
10	SLOW	Make jog slow or fast
11	PICK	Turn on pick cylinder
12	PUL	Turn on pulse
13	VAC	Turn on vacuum
14	ALG	Turn on coil alignment
14	GOTO	Move X & Y Axis to selected point no.
15	HOME	Move X,Y & Z Axis to Origin position
16	SAVE	Save X-Axis & Y-Axis to selected point no.

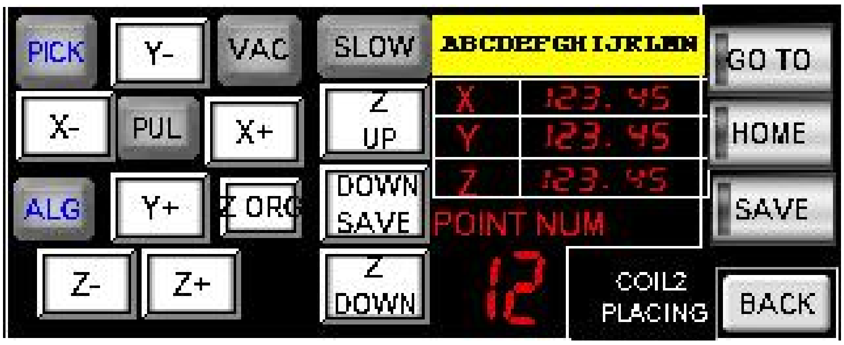


Figure 23 : Coil 2 Placing Screen

COIL 2 PLACING SCREEN

No	Function	Description
1	X+	Move X-Axis to the right
2	X -	Move X-Axis to the left
3	Y+	Move Y-Axis to the frontward
4	Y-	Move Y-Axis to the backward
5	Z+	Move Z-Axis to the downward
6	Z-	Move Z-Axis to the upward
7	Z UP	Move Z-Axis to Z origin point
8	Z DOWN	Move Z-Axis to selected point no.
9	DOWN SAVE	Save Z-Axis to selected point no.
10	SLOW	Make jog slow or fast
11	PICK	Turn on pick cylinder
12	PUL	Turn on pulse

13	VAC	Turn on vacuum
14	ALG	Turn on coil alignment
15	GOTO	Move X & Y Axis to selected point no.
16	HOME	Move X,Y & Z Axis to Origin position
17	SAVE	Save X-Axis & Y-Axis to selected point no.

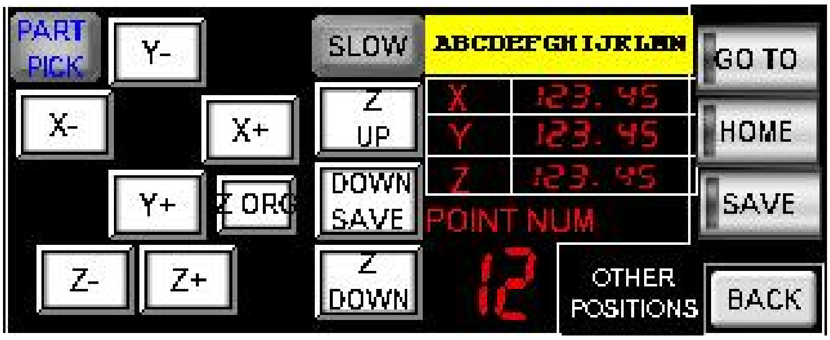


Figure 25 : Other Position Placing Screen

OTHER POSITION PLACING SCREEN

No	Function	Description
1	X+	Move X-Axis to the right
2	X -	Move X-Axis to the left
3	Y+	Move Y-Axis to the frontward
4	Y-	Move Y-Axis to the backward
5	Z+	Move Z-Axis to the downward
6	Z-	Move Z-Axis to the upward
7	Z UP	Move Z-Axis to Z origin point
8	Z DOWN	Move Z-Axis to selected point no.
9	DOWN SAVE	Save Z-Axis to selected point no.
10	SLOW	Make jog slow or fast
11	PART PICK	Turn on Part pick
12	GOTO	Move X & Y Axis to selected point no.
13	HOME	Move X,Y & Z Axis to Origin position
14	SAVE	Save X-Axis & Y-Axis to selected point no.

MACHINE PLC SETTING & LADDER DIAGRAM

0	1	2	3	4	5	
KV-5000	KV-L2*V	KV-B16X*	KV-B16X*	KV-B16R*	KV-B16R*	End Unit
R000 -015		R33000 -33015	R34000 -34015			
R500 -507				R35000 -35015	R36000 -36015	
R30000 -31915	R32000 -32115					

Unit

Select unit(1) | Setup unit(2)

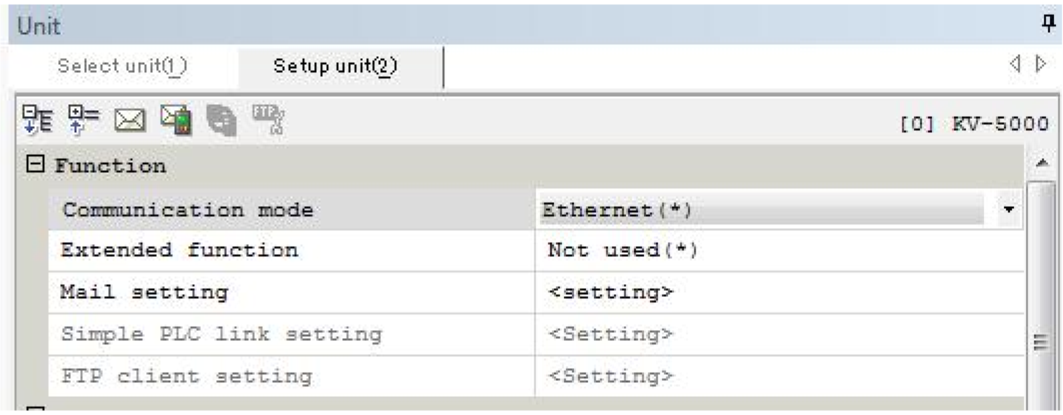
[1] KV-L2*V

Base	
Leading DM No.	DM10100
Number of DMs in use	1
Leading relay No.(set channel ...	R32000
Number of relays in use	32
Node No.	0(*)
Port 1	
Operation mode	KV BUILDER/KV STUDIO mode
Detail	---(*)
Interface	RS-232C(*)
Baudrate	Auto
Data bit length	8 bits(*)
Start bit	1 bits(*)
Stop bit	1 bits(*)
Parity	Even(*)
Checksum	none(*)
RS/CS flow control	OFF(*)
Port 2	
Operation mode	KV BUILDER/KV STUDIO mode
Detail	---(*)
Interface	RS-232C
Baudrate	Auto
Data bit length	8 bits(*)
Start bit	1 bits(*)
Stop bit	1 bits(*)
Parity	Even(*)
Checksum	none(*)
Details setting	
Transfer timeout time (secs)	3

Leading DM No.

To set up the leading No. of data memories (DM) to be used in this unit.

0
1
2
3
4
5
KV-5000
KV-L2*V
KV-B16X*
KV-B16X*
KV-B16R*
KV-B16R*
End Unit
R000 -015
R33000 -33015
R34000 -34015
R500 -507
R35000 -35015
R36000 -36015
R30000 -31915
R32000 -32115



Unit
Select unit(1)
Setup unit(2)
[0] KV-5000
Function
Communication mode
Ethernet(*)
Extended function
Not used(*)
Mail setting
<setting>
Simple PLC link setting
<Setting>
FTP client setting
<Setting>

Unit

Select unit(1) Setup unit(2)

[0] KV-5000

Base

Leading DM No.	DM10000
Number of DMs used	16
Leading relay No. (ch unit set...	R30000
Number of relays used	320
Baud rate	100/10 Mbps, Auto(*)
Setting method of IP address	Fixed IP address(*)
IP address	192.168.0.10
Subnet mask	255.255.255.0
Default gateway	0.0.0.0
Port No.(KVS,KV COM+,DB)	8500
Port No. (host link)	8501
Reception timeout[s].	10
Keep Alive[s]	600
FTP server enabled	disabled(*)
Password	[Unset]
FTP server RUN/PROG switching	Yes(*)
Routing setting	no control(*)
Port No.(VT)	8502
Port No.(System expansion)	8504
DNS Server	0.0.0.0
MC protocol port No.(TCP)	5000
MC protocol Port No. (UDP)	5000
MC protocol code	BIN(*)
Type name code of MC protocol	0
Type name of MC protocol	
Simple PLC link port No. (UDP)	0
Auto adjust clock	Not used(*)
SNTP communication timeout[ms]	60
Clock adjustment time time[h]	0
Clock adjustment time[m]	0
Clock adjustment time[s]	0
Clock adjustment interval[m]	60
GMT offset	+(*)
GMT offset[h]	9
GMT offset[m]	0
GMT offset[s]	0
NTP(SNTP) server	0.0.0.0

⊟ Socket1	
KV Socket	disabled(*)
Byte replacement	H->L(*)
Command sub-header	0060
Response	Without(*)
Response sub-header	E0
Communication direction	Send(*)
⊟ Socket2	
KV Socket	disabled(*)
Byte replacement	H->L(*)
Command sub-header	0060
Response	Without(*)
Response sub-header	E0
Communication direction	Send(*)
⊟ Socket3	
KV Socket	disabled(*)
Byte replacement	H->L(*)
Command sub-header	0060
Response	Without(*)
Response sub-header	E0
Communication direction	Send(*)
⊟ Socket4	
KV Socket	disabled(*)
Byte replacement	H->L(*)
Command sub-header	0060
Response	Without(*)
Response sub-header	E0
Communication direction	Send(*)
⊟ Socket5	
KV Socket	disabled(*)
Byte replacement	H->L(*)
Command sub-header	0060
Response	Without(*)
Response sub-header	E0
Communication direction	Send(*)
⊟ Socket6	
KV Socket	disabled(*)
Byte replacement	H->L(*)
Command sub-header	0060
Response	Without(*)
Response sub-header	E0
Communication direction	Send(*)
⊟ Socket7	
KV Socket	disabled(*)
Byte replacement	H->L(*)
Command sub-header	0060
Response	Without(*)
Response sub-header	E0
Communication direction	Send(*)
⊟ Common KV socket	
Response timeout[s]	30

n

Introduction

This section describes the features and specifications of the Is Auto Coil Machine.

About IS Auto Coil Machine

The purpose of the system is to assembly , the base with 2 pcs of solenoid coil and apply glue with bonding.

- Simple operation

  The system is designed and emphasizes on simple, easy teaching and automatic operations.

- Modern and user-friendly user interface

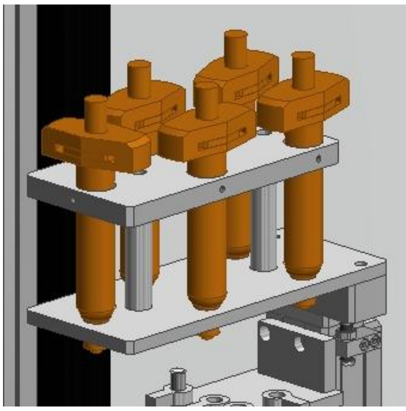
  Simple and modern GUI of the controlling software features many functions and even easy to use.

System Specifications

The system specification will be listed below

Mechanical and Electrical Specifications

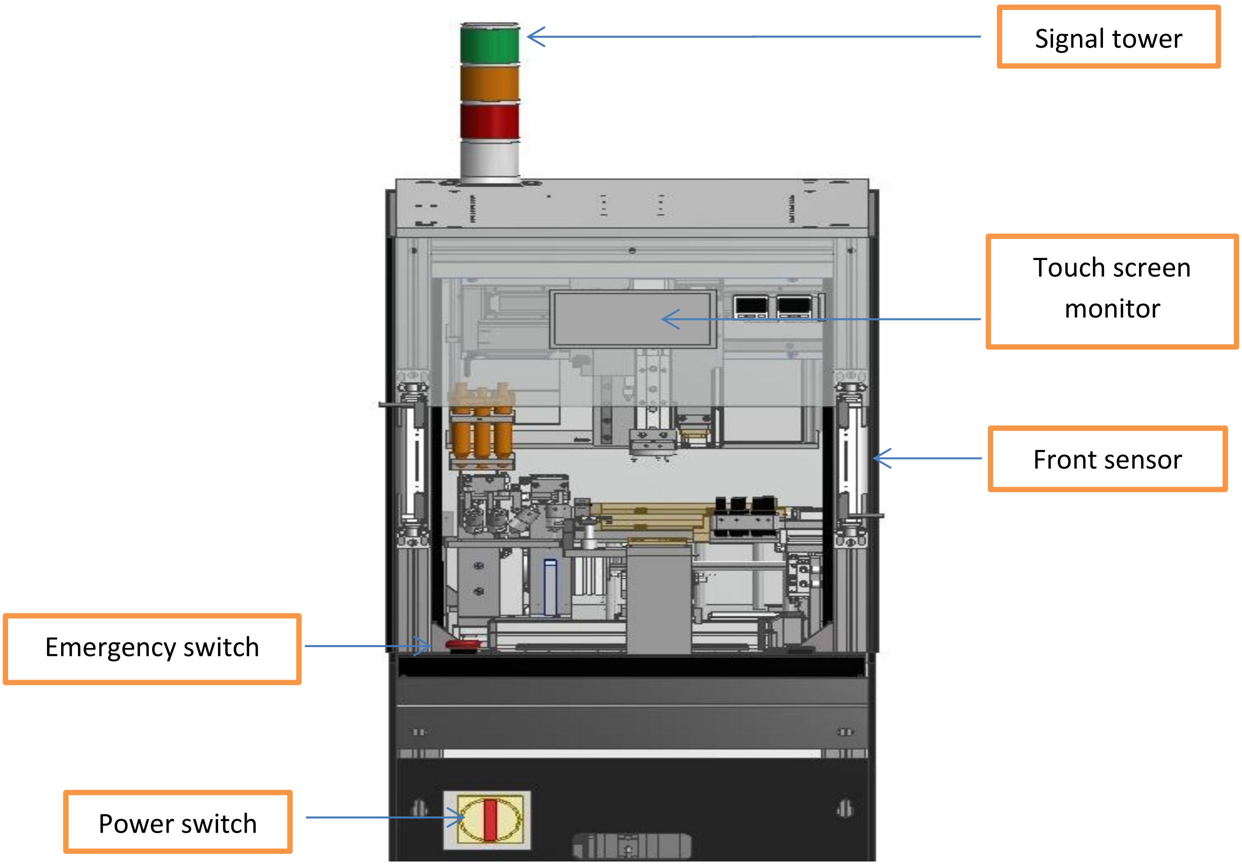
Item	Specifications
Control PLC	Keyence PLC
Power supply	100V AV - 110V AC
Air pressure	Min 0.5 Mpa or 5 bar (dry air)
Cycle time/pcs	49s/pcs



Names and functions of components

This section main describes names and the functions of the various parts of Is Auto Coil Machine.

External View



- Emergency switch

By pressing the emergency switch will bring the machineto emergency halt status.

- Monitor and touch screen

The touch screen monitor is used for machine operations control.

- Front curtain sensor

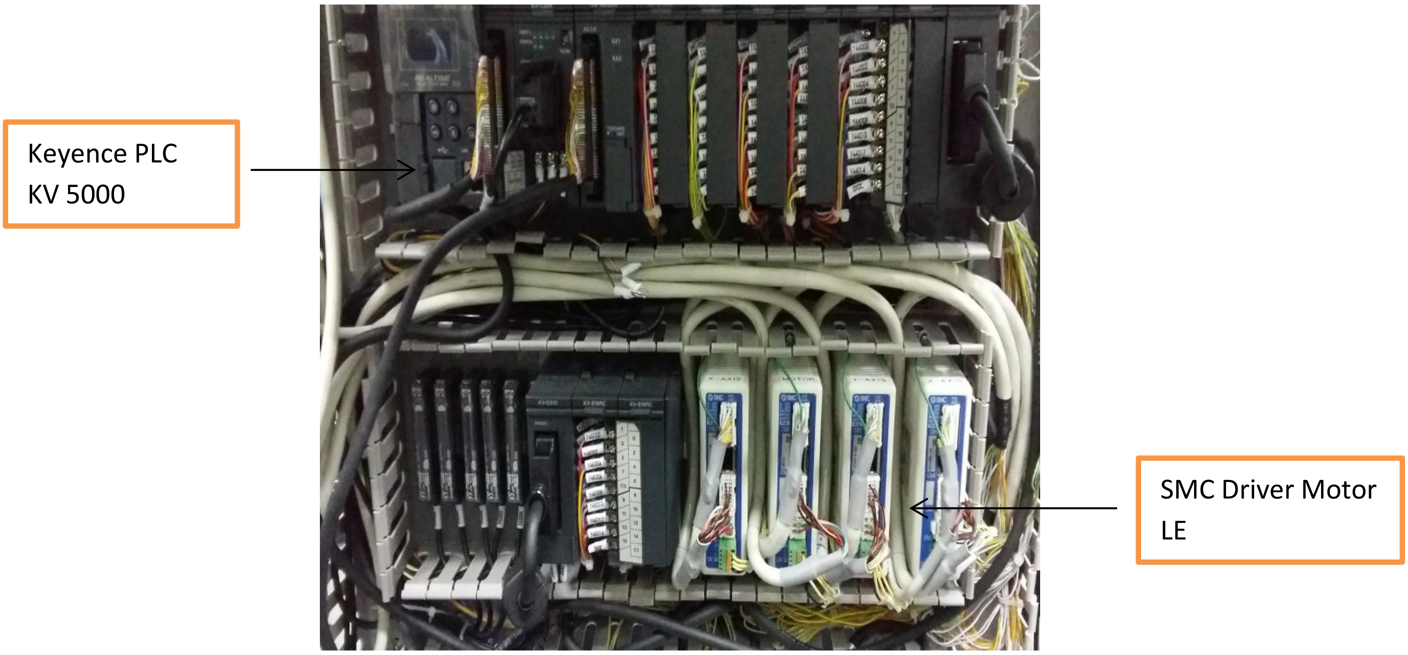
It is a safety prevention tool, only enabled when performing sorting operation. The operation will be halted if the sensor is **ON.**

- Signal tower

It shows the system current status with the green, red and yellow lights with buzzer.

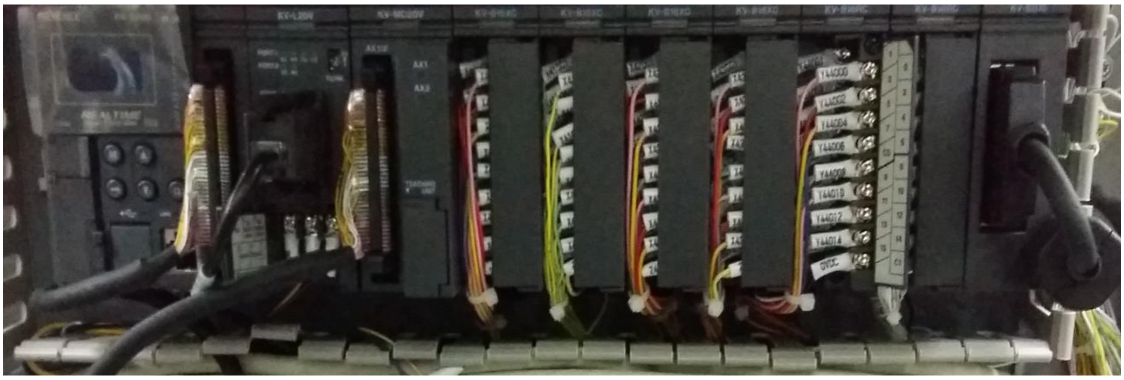
Inside Door

Power Box

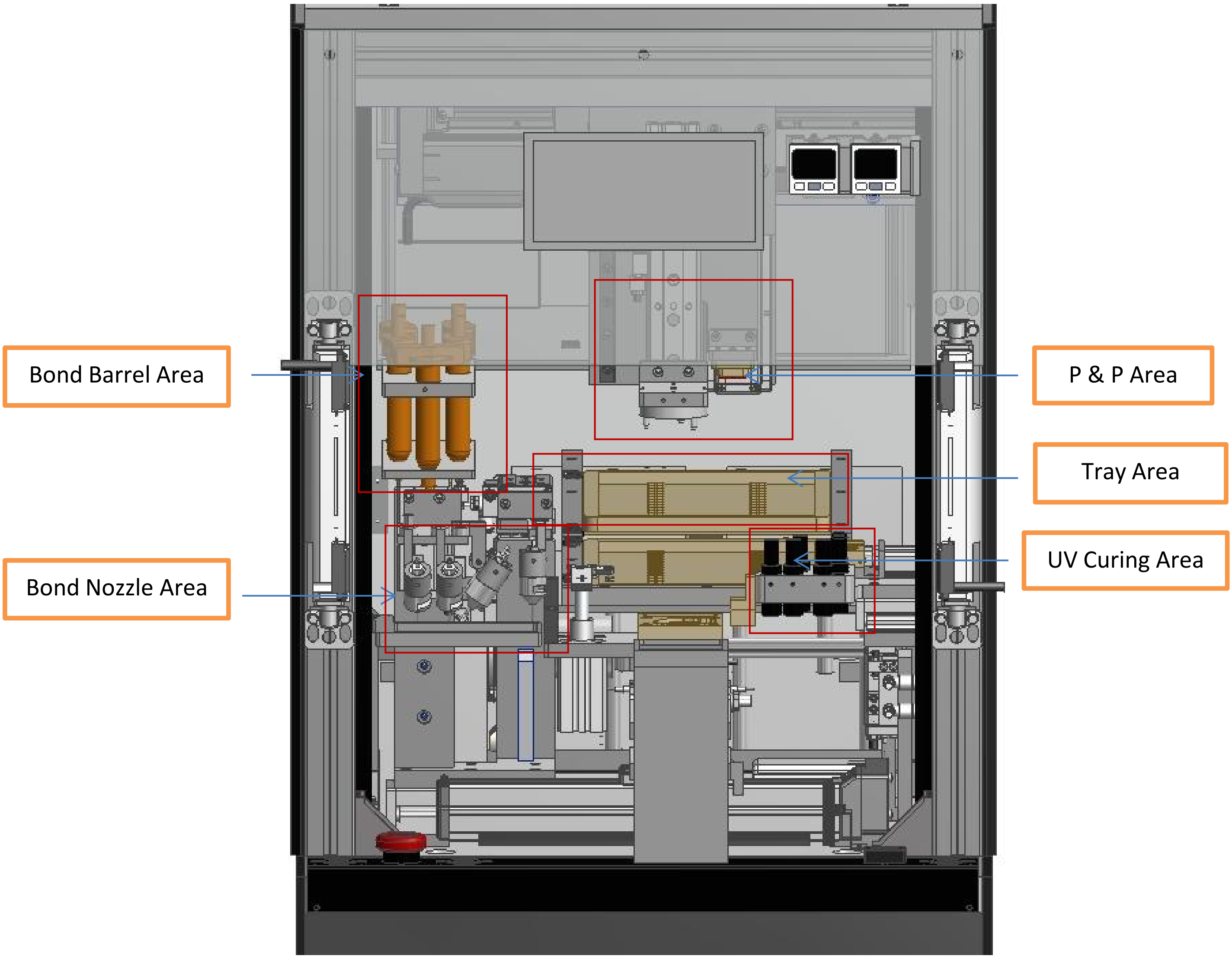


- SMC Servo Motor LE

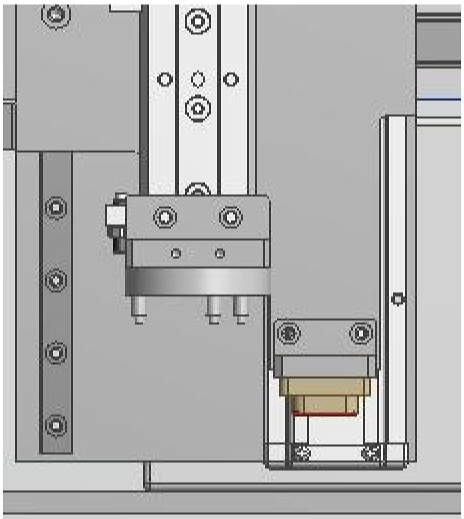
- Keyence PLC KV 5000



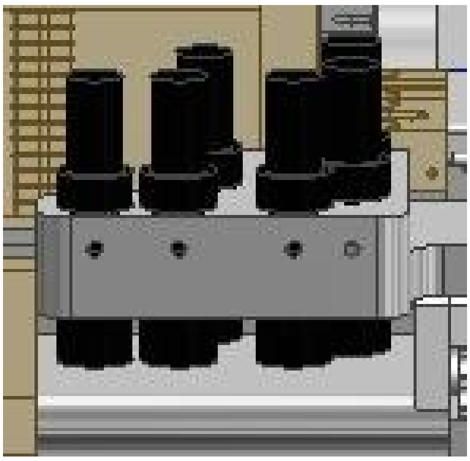
Working Section



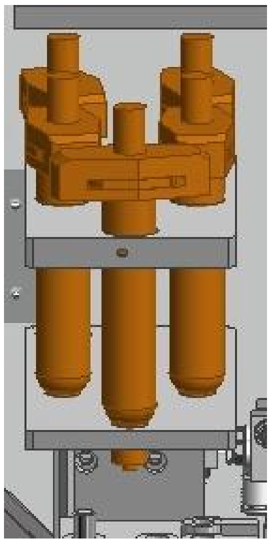
P & P Area



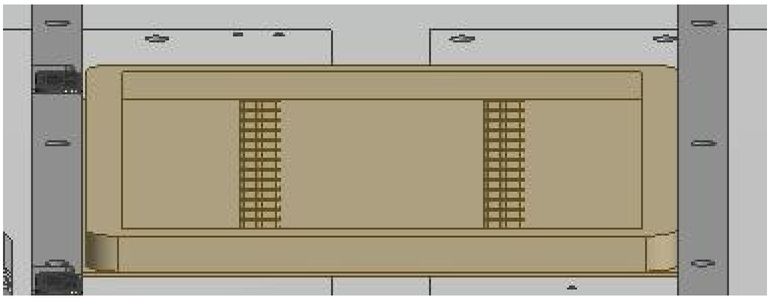
UV Curing Area



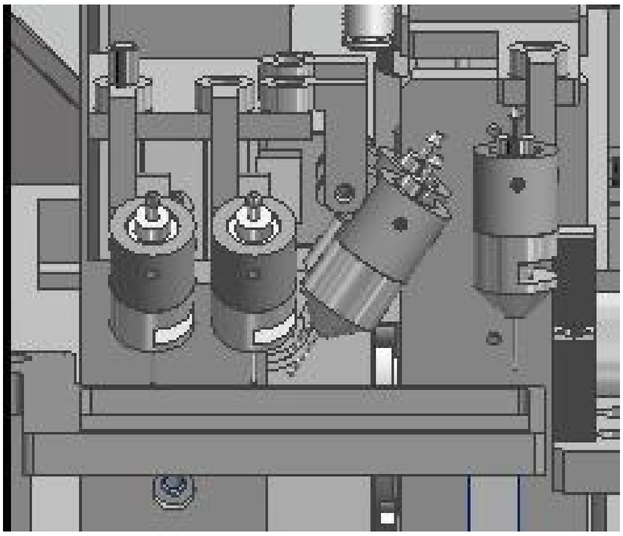
Bond Barrel Area



Tray Area



Bond Nozzle Area



OPERATION MANUAL

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WIRING DIAGRAM

PART LIST

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L273 IS AUTO COIL MACHINE

OPERATION MANUAL